Assignment 5: LED Blinking Circuit Using Arduino

Internship: IoT with AI

Week: 1

Topic: Introduction to IoT and Electronics Basics

1. Introduction

Arduino is an open-source microcontroller platform used to develop electronic and IoT projects. It can control electronic components such as LEDs, motors and sensors through programming.

In this experiment, three LEDs (red, yellow and green) are connected to an Arduino Uno to demonstrate sequential LED blinking, similar to a traffic light system.

2. Objective

To design and simulate an LED blinking circuit using Arduino Uno and understand how digital output pins can control multiple LEDs with specific time delays.

3. Components Required

1. Arduino Uno R3 – 1
2. Breadboard – 1
3. Red LED – 1
4. Yellow LED – 1
5. Green LED – 1
6. 220Ω resistors – 3
7. Jumper wires – As required
8. Tinkercad Circuits – Online simulation platform

4. Circuit Connections

The three LEDs are connected to different digital pins of Arduino Uno.

- Red LED is connected to digital pin 13.
- Yellow LED is connected to digital pin 12.
- Green LED is connected to digital pin 8.
- Each LED is connected through a 220Ω current-limiting resistor.
- The negative terminals of the LEDs are connected to Arduino GND through their respective resistors.

5. Working Principle

The Arduino controls the LEDs by sending HIGH and LOW signals through its digital output pins.

Initially, the red LED turns ON for 2 seconds while the other LEDs remain OFF.

Next, the yellow LED turns ON for 1 second.

Finally, the green LED turns ON for 2 seconds.

The sequence repeats continuously, demonstrating a basic traffic light control system.

6. Arduino Program

```
int redLED = 13;
int yellowLED = 12;
int greenLED = 8;

void setup() {
  pinMode(redLED, OUTPUT);
  pinMode(yellowLED, OUTPUT);
  pinMode(greenLED, OUTPUT);
}
```

```
void loop() {
  digitalWrite(redLED, HIGH);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, LOW);
  delay(2000);

  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, HIGH);
  digitalWrite(greenLED, LOW);
  delay(1000);

  digitalWrite(redLED, LOW);
  digitalWrite(yellowLED, LOW);
  digitalWrite(greenLED, HIGH);
  delay(2000);
}
```

7. Simulation Procedure

1. Open Tinkercad Circuits and create a new circuit.
2. Add Arduino Uno and a breadboard.
3. Place three LEDs on the breadboard.
4. Connect the LEDs to Arduino pins 13, 12 and 8.
5. Connect a 220Ω resistor in series with each LED.
6. Connect the LED circuits to Arduino GND.
7. Open the Code editor and select Text mode.
8. Enter the Arduino program.
9. Click Start Simulation.
10. Observe the LEDs glowing sequentially.

8. Expected Output

- Red LED turns ON for 2 seconds.
- Yellow LED turns ON for 1 second.
- Green LED turns ON for 2 seconds.
- The sequence repeats continuously.

9. Applications

1. Traffic light control systems.
2. LED indication systems.
3. Electronic signalling systems.
4. Basic automation projects.
5. Embedded systems learning.

10. Conclusion

The LED blinking circuit demonstrates how Arduino Uno can control multiple LEDs using digital output pins. The experiment provides practical understanding of Arduino programming, digital signals, circuit connections and timing functions. It also introduces the basic working principle of traffic light control systems.